CSE5026: Introduction and Frontiers of Cognitive Science

Language Modeling

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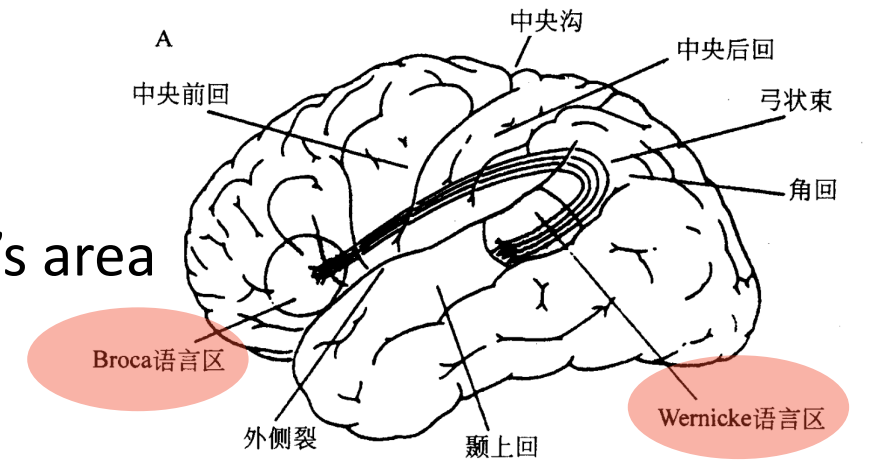
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Overview

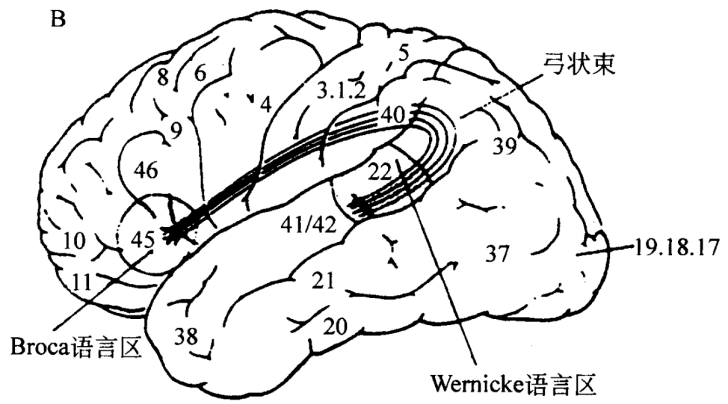
- Language Modeling（语言建模)
- Cognitive Effort in Language Processing（语言加工中的认知负载)
- Dialogue

Review: Brain and Language Comprehension



Broca's area

Wernicke's area

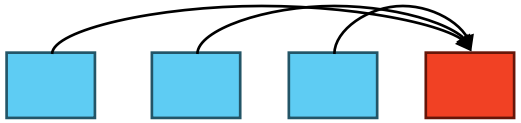


white matter pathways

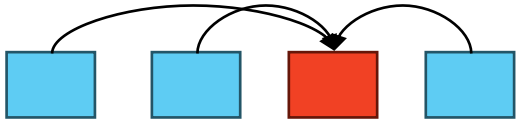
- 声音信号在听觉系统被转换，信息传递到以角回为中心的顶颞枕联合皮质，再传递到 **Wernicke**区，在这里从语音信息中提取出单词表征 (word representation)。
- 信息流从 Wernicke 区经过弓形束(白质神经束)到达 Broca 区，这里是语法特征 (syntactic representation) 记忆之所。
- 接着单词表征激活概念中心相关的概念 (concept)。
- 听觉理解 (comprehension) 就发生了。

Language Model

- LM: A computational model of predicting words: $w_1, w_2, \ldots, w_N$
- Auto regressive LM: $p(w_i|w_1, \ldots, w_{i-1})$ => $p(w_i|w_{<i})$



- Masked LM: $p(w_i|w_1, \ldots, \boldsymbol{w_{i-1}}, \boldsymbol{w_{i+1}}, \ldots, w_N)$ => $p(w_i|w_{\neq i})$



Language Model

- Early: N-gram language models.
- Transformer-based Auto Regressive LMs.
- Diffusion-based LMs.

How to Train an LM?

- Pre- Neural network solution: ***n*-gram Language Model**
- **Def.** An *n*-gram is a chunk of *n* consecutive words

```
the Sun rises every _____
```

- **Uni**grams (*n*=1): “the”, “Sun”, “rises”, “every”
- **Bi**grams (*n*=2): “the Sun”, “Sun rises”, “rises every”
- **Tri**grams (*n*=3): “the Sun rises”, “Sun rises every”
- **Four**-grams (*n*=4): “the Sun rises every”

- **Idea**: Count the frequencies of different n-grams and use these to predict the next word

n-grams LM: Markov assumption



Andrey Andreyevich Markov
(14 June 1856 – 20 July 1922)

- **Markov assumption**: a word at only depends on its preceding $n-1$ words

$$P\left(x^{\langle t+1\rangle} \middle| x^{\langle 1\rangle}, \ldots, x^{\langle t\rangle}\right) = P\left(x^{\langle t+1\rangle} \middle| \underbrace{x^{\langle t-n+2\rangle}, \ldots, x^{\langle t\rangle}}_{n-1 \text{ words}}\right)$$

By the definition of conditional probability

Probability of a *n*-gram

Probability of a (*n-1*)-gram

$$= \frac{P\left(x^{\langle t-n+2\rangle}, \ldots x^{\langle t\rangle}, x^{\langle t+1\rangle}\right)}{P\left(x^{\langle t-n+2\rangle}, \ldots x^{\langle t\rangle}\right)}$$

- **Question**: How to obtain the probabilities?
- **Answer**: By counting them from some large enough corpora (statistical approximation)

$$\approx \frac{\text{count}\left(x^{\langle t-n+2\rangle}, \ldots x^{\langle t\rangle}, x^{\langle t+1\rangle}\right)}{\text{count}\left(x^{\langle t-n+2\rangle}, \ldots x^{\langle t\rangle}\right)}$$

n-gram LM: Example

- Goal: Learning a **4**-gram LM, i.e., considering 3 preceding words

discard

~~in this peculiar game~~, the Sun rises every w

$$P(w|\texttt{Sun rises every}) \approx \frac{\text{count}(\texttt{Sun rises every } w)}{\text{count}(\texttt{Sun rises every})}$$

Example, suppose in the corpus

- "Sun rises every" occurs **1000** times
- "Sun rises every morning" occurs **600** times
  - $\Rightarrow P(\textbf{morning}|\texttt{Sun rises every}) = \mathbf{0.6}$
- "Sun rises every day" occurs **300** times
  - $\Rightarrow P(\textbf{day}|\texttt{Sun rises every}) = \mathbf{0.3}$

Question: What's the problem of this method?

Sparsity Problem with *n*-gram LM

$$P(w|\texttt{Sun rises every}) \approx \frac{\text{count}(\texttt{Sun rises every } w)}{\text{count}(\texttt{Sun rises every})}$$

Sparsity problem 1:
What if "Sun `rises every` w" never occurred in data? Then the probability is 0

Partial solution: **Smoothing** => Add small δ to the count for every word in V

Sparsity problem 2:
What if "Sun `rises every`" never occurred in data? Then the probability is not computable

Partial solution: **Backoff** => Count "`rises every`" instead, i.e., shorter conditional context

Adapted from: https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1224/

Storage Problem with *n*-gram LM

- **Storage**: Need to store count for all *n*-grams in the corpus
- Larger *n* or larger corpus means larger model size

$$P(w|\texttt{Sun rises every}) \approx \frac{\texttt{count(Sun rises every } w\texttt{)}}{\texttt{count(Sun rises every)}}$$

Every term needs be stored

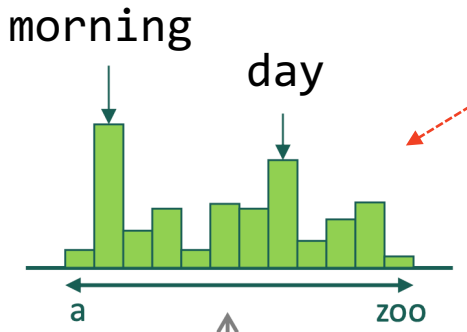
Fixed-Window Neural Language Model

- **Idea**: Represent words with embedding vectors
- predict the next word using the concatenated embeddings from a fixed context window

output

$$\hat{\boldsymbol{y}} = \text{softmax}(U\boldsymbol{h}^\top + b_2)$$

dim: $|V|$ $\quad$ $|V|\times d'$



U

hidden layer

$$\boldsymbol{h} = g(W\boldsymbol{e}^\top + b_1)$$

dim: d' $\quad$ $d'\times 4d$

W

concatenated word embeddings

$$\boldsymbol{e} = \underbrace{\left[\boldsymbol{e}^{\langle 1\rangle}; \boldsymbol{e}^{\langle 2\rangle}; \boldsymbol{e}^{\langle 3\rangle}; \boldsymbol{e}^{\langle 4\rangle}\right]}_{4\times d} \} d$$

Input tokens: $x^{\langle 1\rangle}, x^{\langle 2\rangle}, x^{\langle 3\rangle}, x^{\langle 4\rangle}$

(window size = 4)

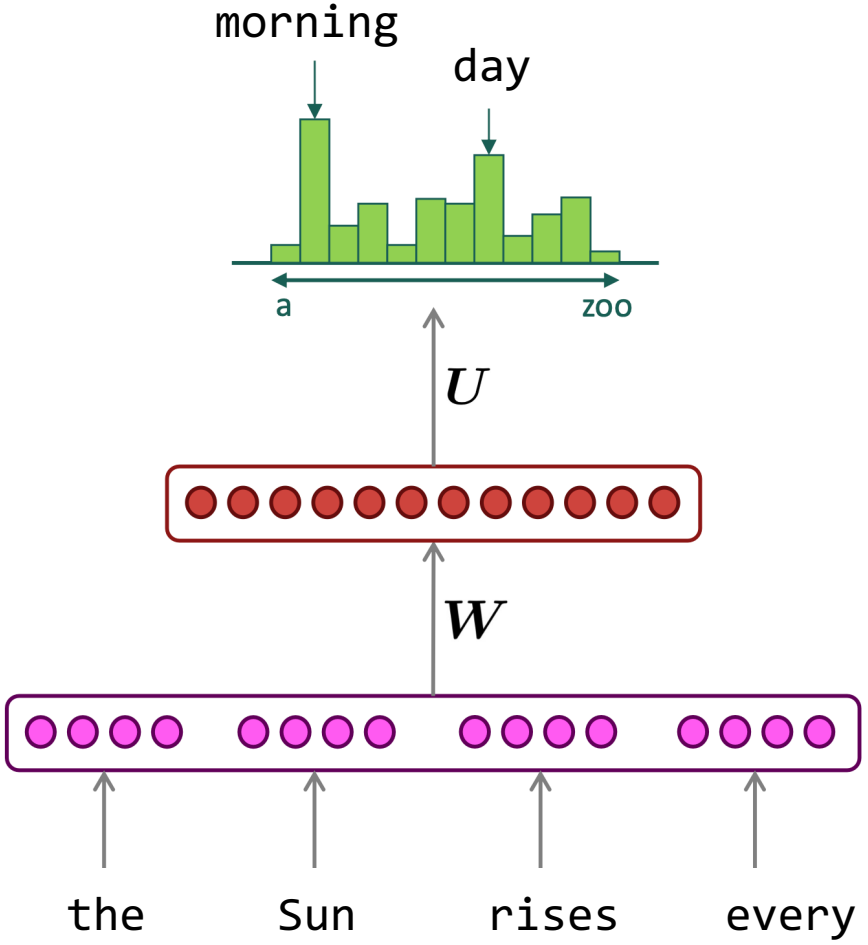
the Sun rises every

Figure from: https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1224/

Fixed-Window Neural LM

- **Improvements** over n-gram LM:
- No sparsity problem;
- No need to store all observed n-grams

- Remaining **problems**:
- Fixed window can be small
- Increasing window size also increase W
- Need a neural architecture that can process *any input length*



Overview

- Language Model
- **Cognitive Effort in Processing Language (语言加工中的认知负载)**
- Dialogue

Information changes when word *omitted*

- s1: *My boss confirmed **that** we were absolutely crazy.*
- *s1: *My boss confirmed **we** were absolutely crazy.*

- Which sentence reads “smoother”?

$$p(\textbf{that} \mid My\ boss\ confirmed) > p(\textbf{we} \mid My\ boss\ confirmed)$$

Measuring Information in Language

- Surprisal (惊奇度) = negative log-probability (NLL)
- Alternatively,
  - Information density
  - Entropy (not strict)

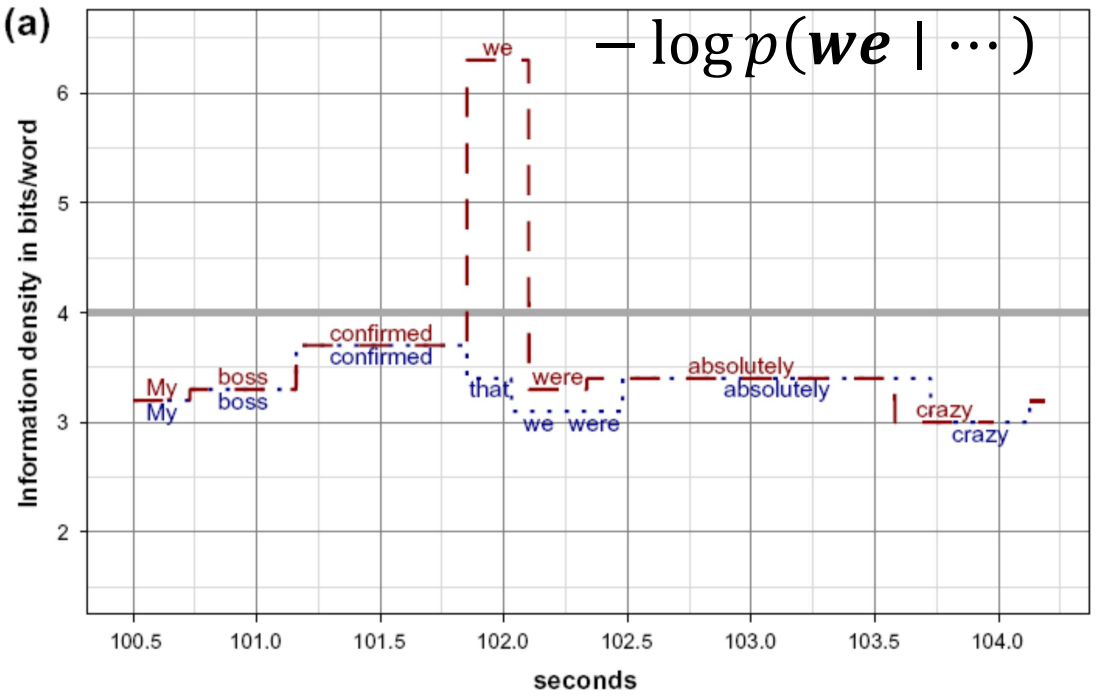
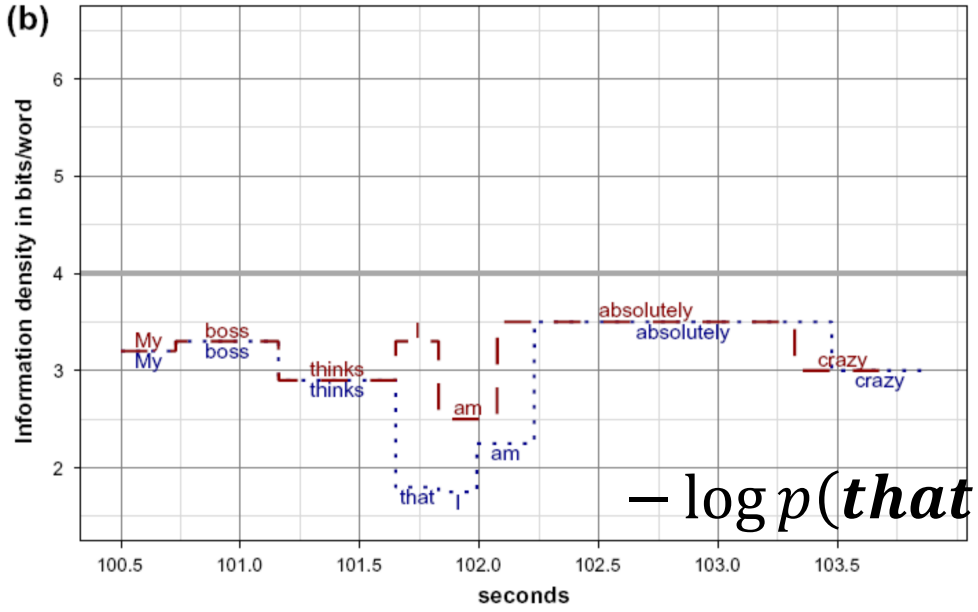


Fig. from Jaeger et al. (2010), http://doi.org/10.1016/j.cogpsych.2010.02.002

Information changes when words *added*

- s2: *My boss thinks **I** am absolutely crazy.*
- *s2: *My boss thinks **that** I am absolutely crazy.*



$$-\log p(\boldsymbol{that} \mid \cdots)$$

$$p(\boldsymbol{that} \mid My\ boss\ thinks) < p(\boldsymbol{I} \mid My\ boss\ thinks)$$

Fig. from Jaeger et al. (2010), http://doi.org/10.1016/j.cogpsych.2010.02.002

Uniform Information Density

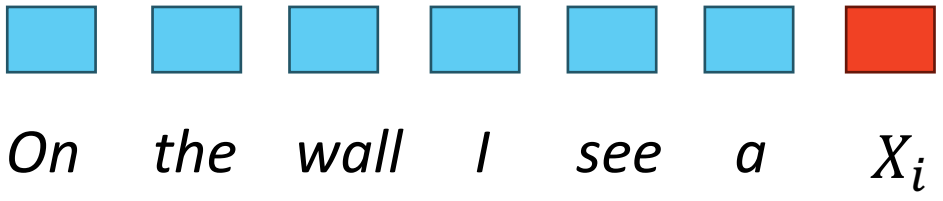
- Uniform Information Density (UID) predicts that language production is affected by a (human) preference to distribute information ***uniformly*** across the linguistic unit.

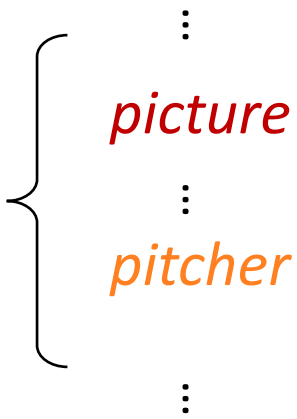
UID is a complex problem

- s1: 我 买了 三 张 桌子
  I bought three CL.flat tables
- *s1: 我 买了 三 个 桌子
  I bought three CL.general tables
- CL = Classifier (量词), CL.flat (特殊量词, specific classifier)
  CL.general (通用量词, general classifier)
- Question: When we decide the classifier, what words are taken into as context?

Source: Zhan et al. (2018), https://aclanthology.org/N18-1181/

LM tells about Information in Language





X_i can have $|V|$ outcomes, among which

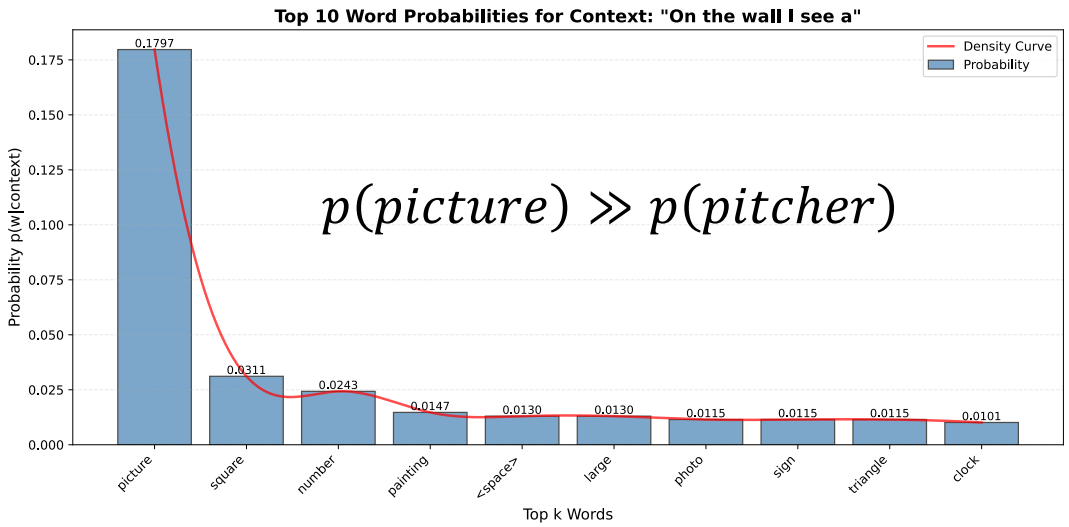
$p(X_i = picture|C) = 0.179290$

$p(X_i = pitcher|C) = 6.6\times10^{-5}$

(according to qwen3-9.6B-base)

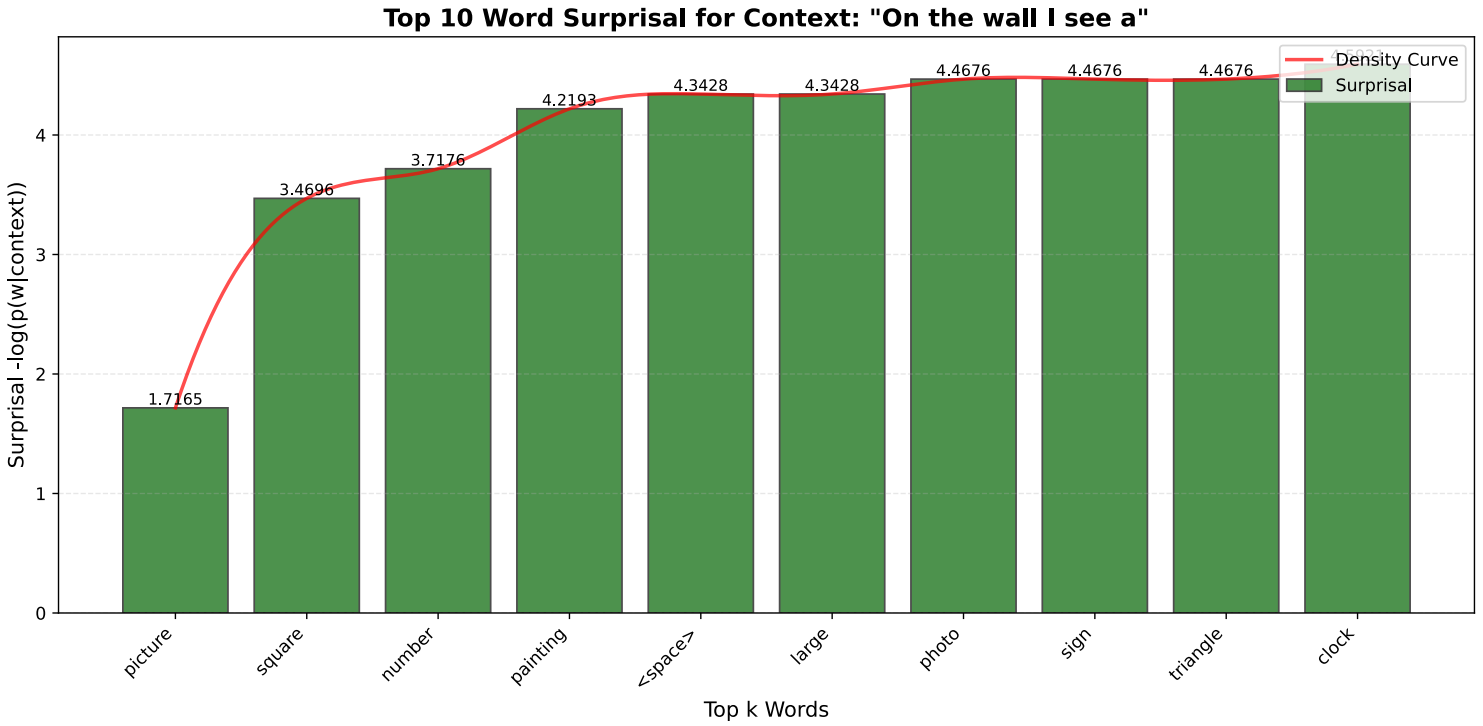
Let a random variable X_i describe the event of generating a word at position i, given its context C.

"*picture*" is a much more likely outcome of the event



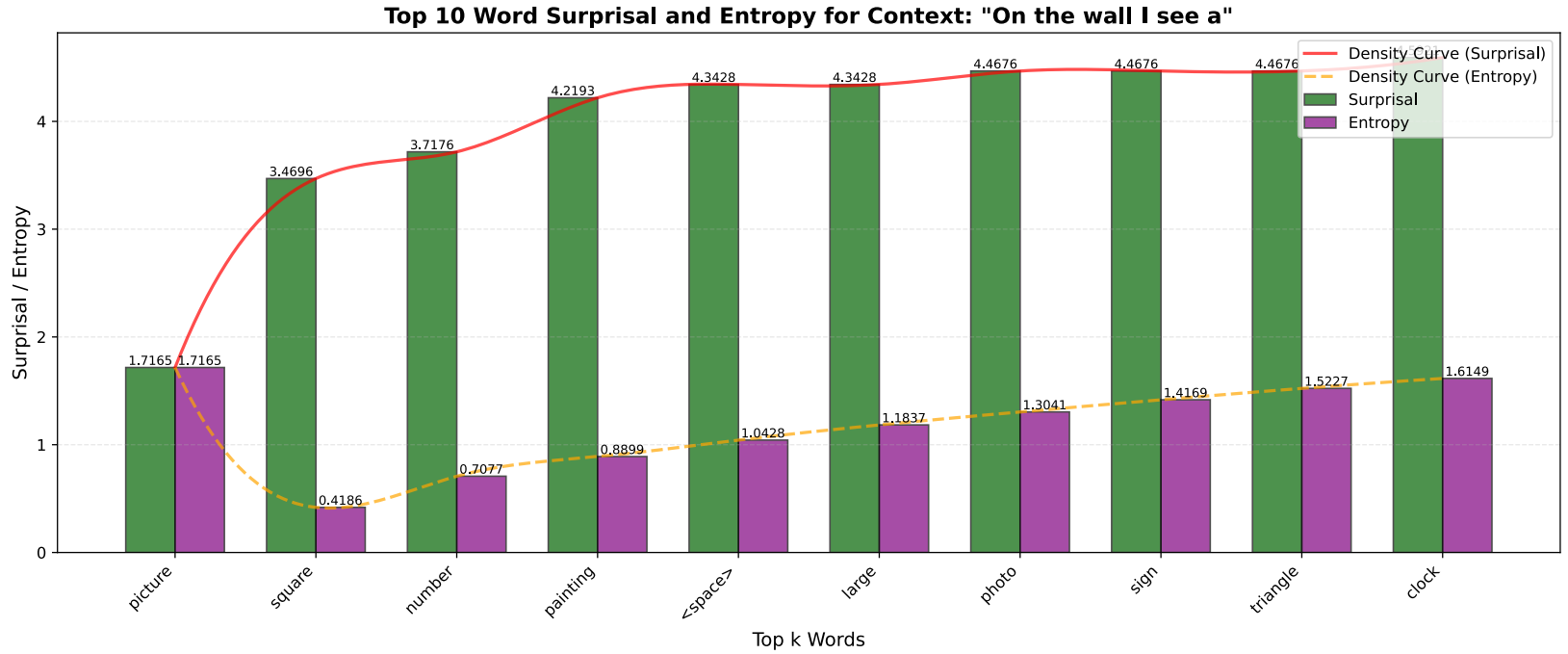
Surprisal of Event Outcome

- Surprisal of an event outcome: $s(x_j) = -\log p(x_j|C)$
  - $s(picture) = -\log p(picture|C) = 1.719$, $s(pitcher) = -\log p(pitcher|C) = 9.625$



Entropy of Random Variable (Shannon's entropy)

- Entropy of rand. var. X_i, total amount of uncertainty measured by weighted sum of surprisal: $H(X_i) = -p_j \sum_{j=1}^{|V|} \log p(x_j|C)$

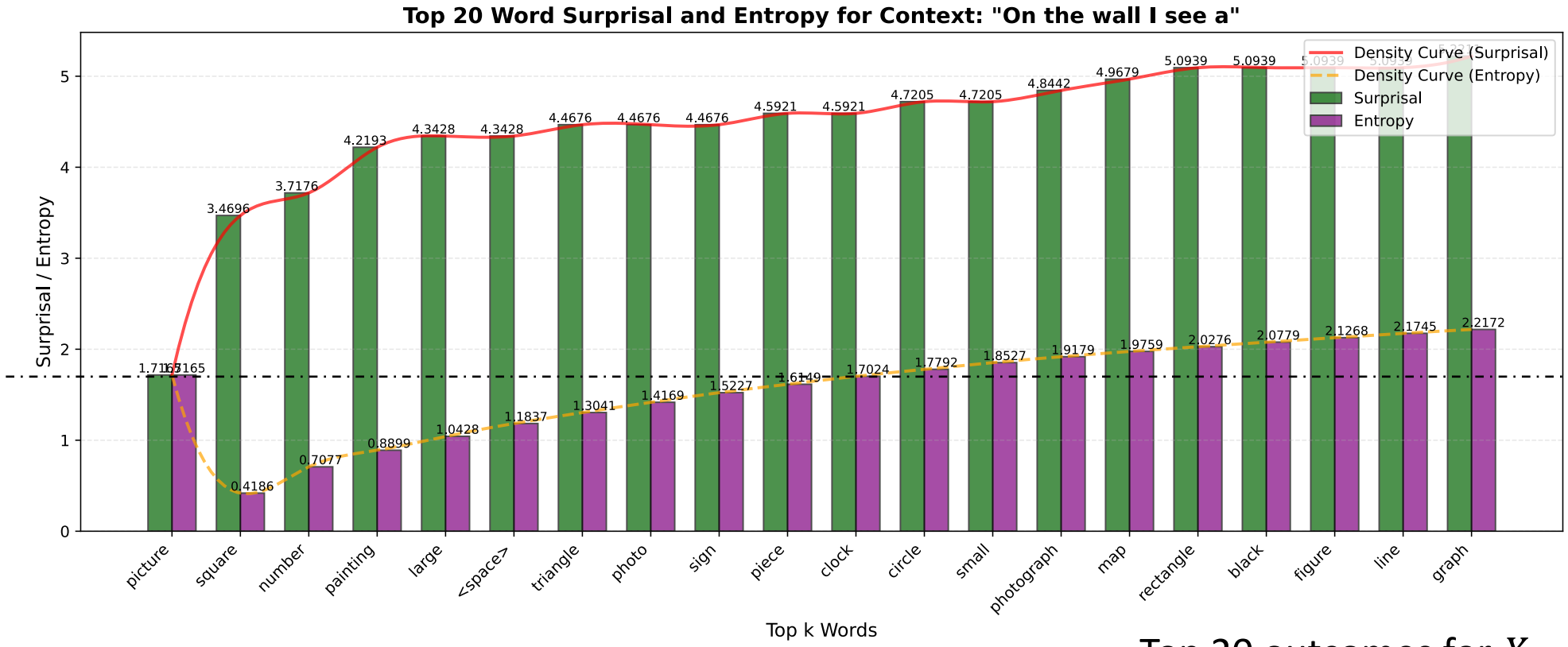


Purple bar (dashed) shows an inaccurate estimation of entropy using the top-k words:

$$H' = -p_j \sum_{j=1}^{k} \log p_j$$

Entropy of Random Variable

- The empirical values of entropy H are larger than surprisal s



H' goes up

$s(picture)$

Top 20 outcomes for X_i

Surprisal can approximate entropy

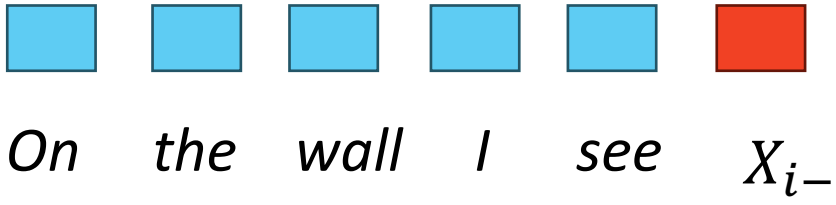
- Entropy measures the expected surprisal of an unknown event X_i



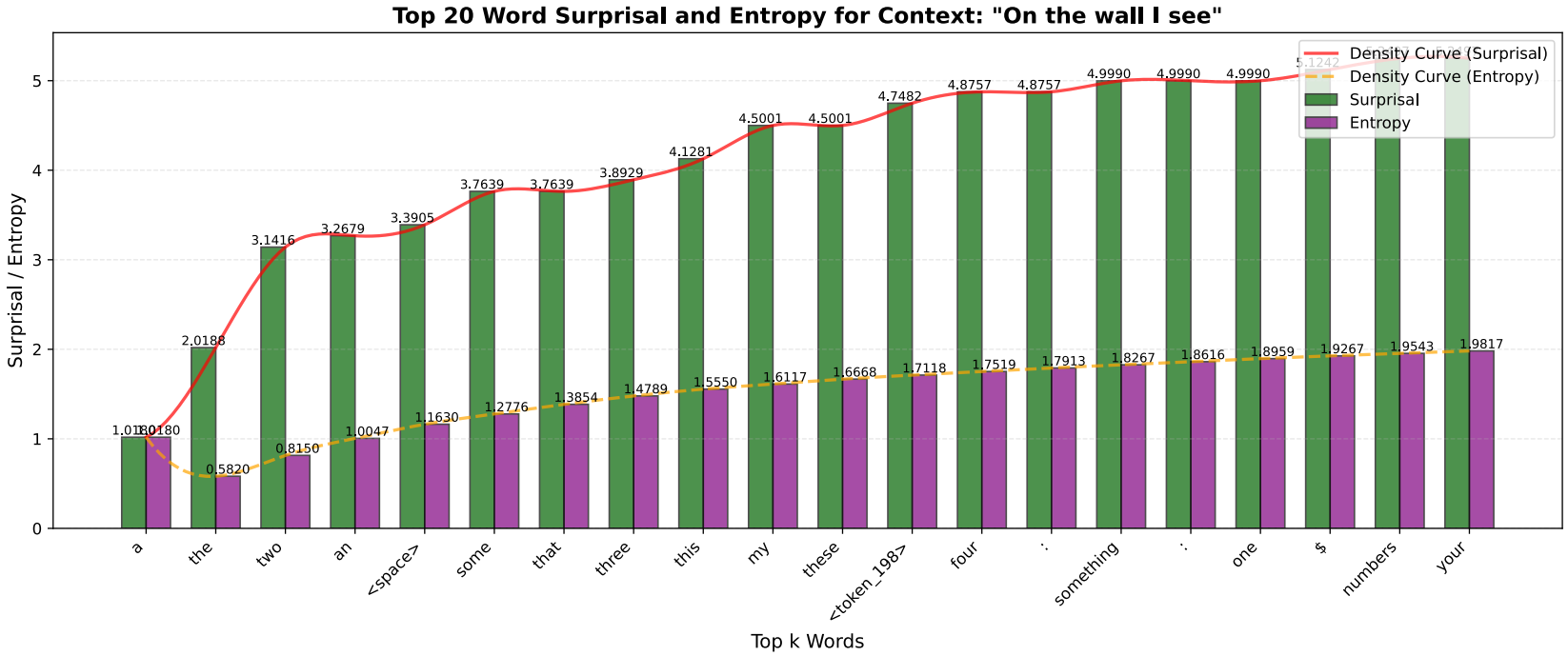
$$H = -p_j \sum_{j=1}^{|V|} \log p_j$$

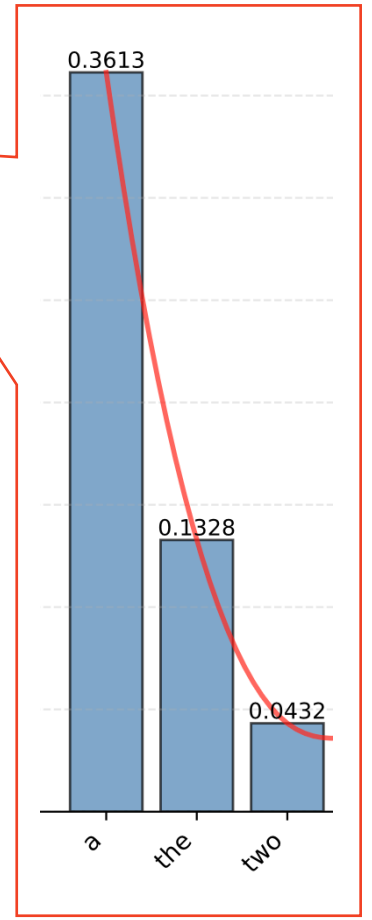
- Surprisal measures the surprisal of a known outcome: $s(x_j) = -\log p(x_j|C)$
- In natural language, the amount of "information" can be defined differently.

Compute entropy for different positions



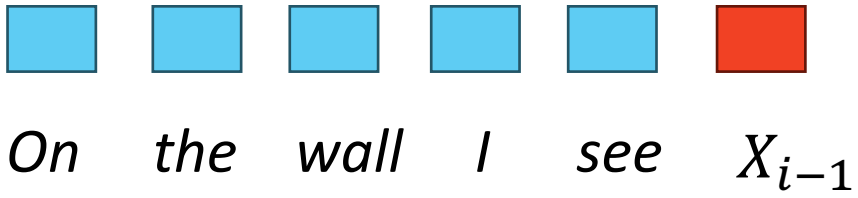
$$\begin{cases} p(X_{i-1} = a|C) = 0.3613 \\ p(X_{i-1} = the|C) = 0.1328 \\ \vdots \end{cases}$$





Measure Information with Entropy Reduction

- Entropy reduction: How much uncertainty is reduced due to the occurrence of a word?

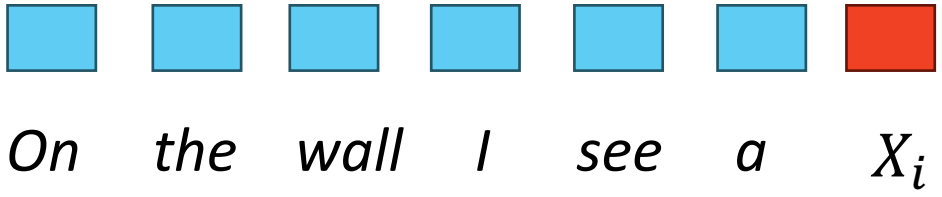


$H'(X_{i-1}) \approx 1.9817$
using $k = 20$



$\Delta H = H'(X_i) - H'(X_{i-1}) = -0.2355$



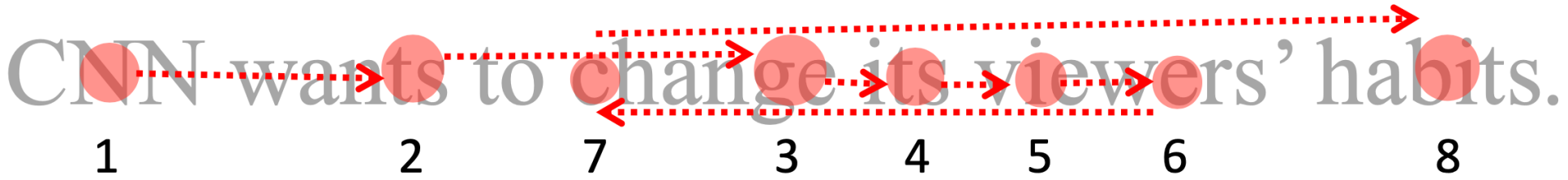


$H'(X_i) \approx 2.2172$

Negative reduction means the uncertainty increases due to the production of “a”.

Source: https://acl2025-eyetracking-and-nlp.github.io/

Eye-movement: Measure of Cognitive Effort



Fixation 注视



Saccades /sæˈkɑːd/ 扫视

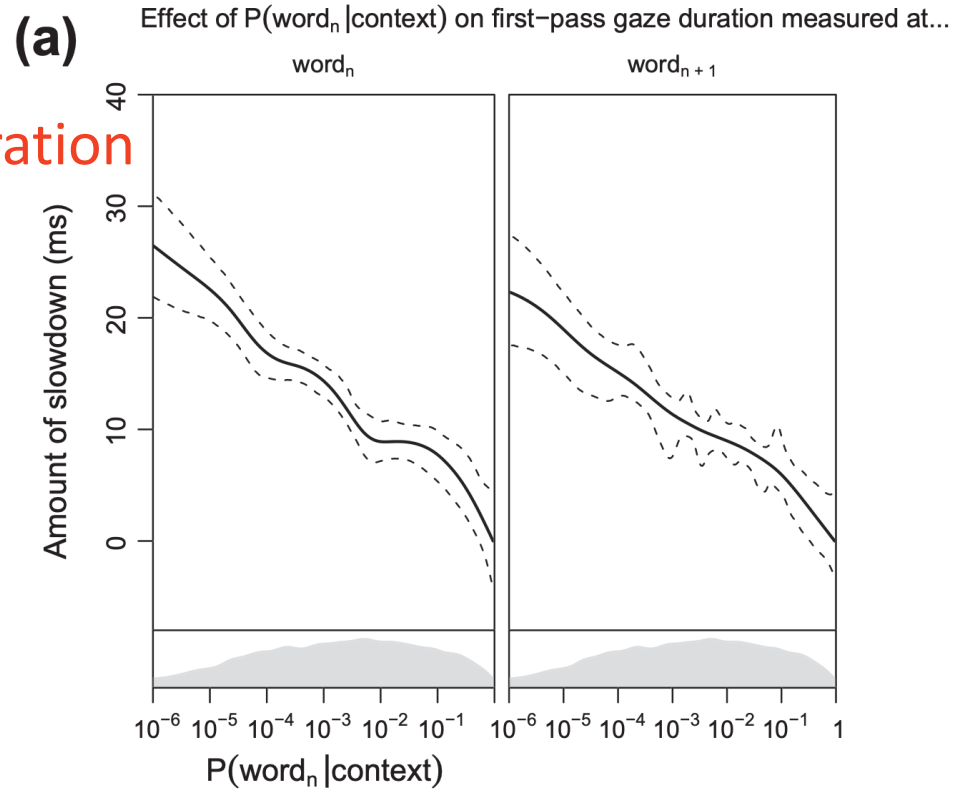


Source: Smith and Levy, 2013, http://dx.doi.org/10.1016/j.bandl.2014.10.006

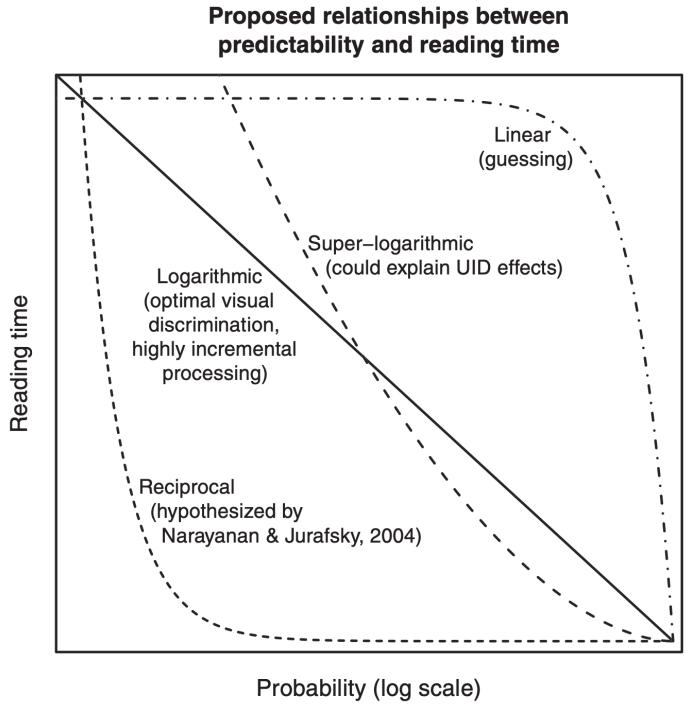
Information and Cognitive Effort

- Words of lower probability take more time to read

Gaze duration



from n-gram LM



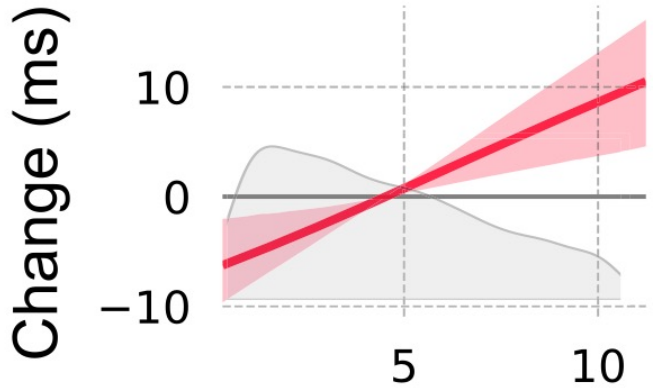
Several different theories to explain the relationship

Source: https://doi.org/10.1073/pnas.2307876121

Information and Cognitive Effort

- Word predictability has *logarithmic* effect on reading time.
  - lower probability p => higher surprisal $-\log p$

Relative reading time



$-\log p$ from GPT-2, GPT-J, GPT-3 etc.

Source: https://acl2025-eyetracking-and-nlp.github.io/

Are LLMs aligned with Human Reading?

- A research direction: Predictive power of LLM-extracted surprisal (or other metrics) for human reading times (RTs)

Step 1:
Fit two regression models predicting reading times, **with** and **without surprisal as predictor**

Step 2:
Compute Log-Likelihood (LL) of each model

Step 3:
Predictive Power (PP) of surprisal is the **ΔLL** of the two models

$$\mathcal{M}_{baseline}: RT \sim baseline_variables \longrightarrow LL(\mathcal{M}_{baseline})$$

$$\mathcal{M}_{surprisal}: RT \sim baseline_variables \oplus surprisal \longrightarrow LL(\mathcal{M}_{surprisal})$$

$$\longrightarrow \text{PP(surprisal)} := \Delta LL(\mathcal{M}_{surprisal}, \mathcal{M}_{baseline}) := LL(\mathcal{M}_{surprisal}) - LL(\mathcal{M}_{baseline})$$

Source: https://acl2025-eyetracking-and-nlp.github.io/

Findings and Open Space

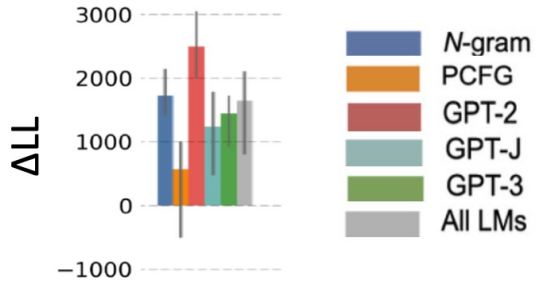
Older Models

Better LM --> surprisal predicts RTs better

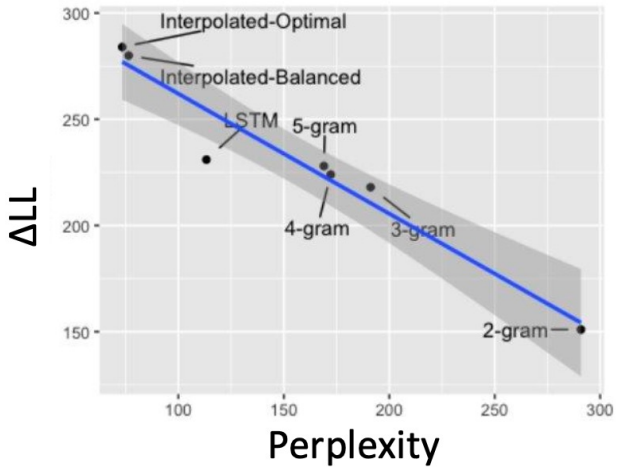
LLMs

Better LM --> surprisal predicts RTs worse

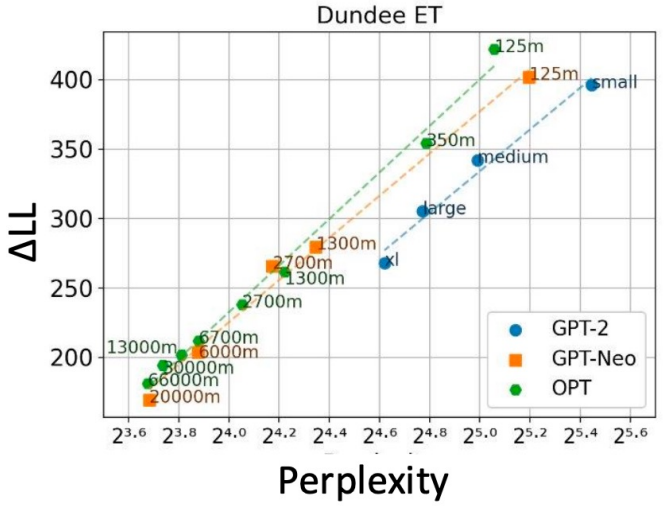
Shain et al. (2023)



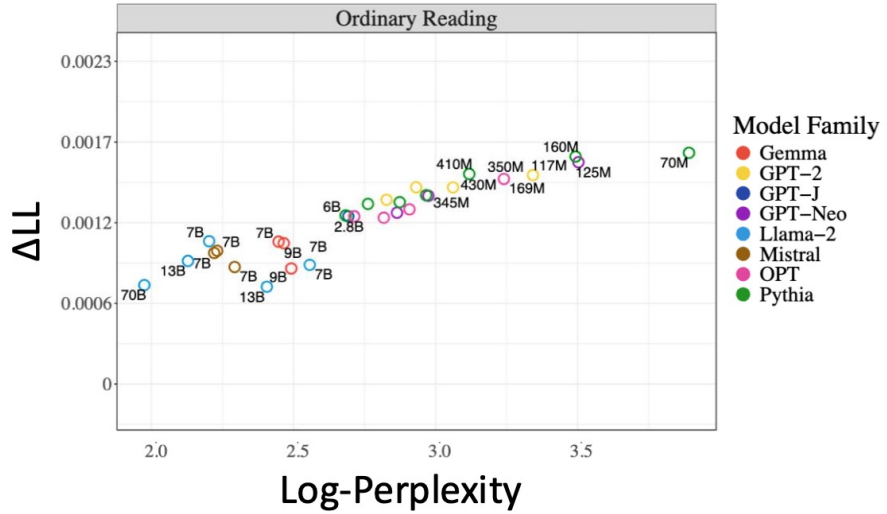
Goodkind and Bicknell (2018)



Oh and Schuler (2023)



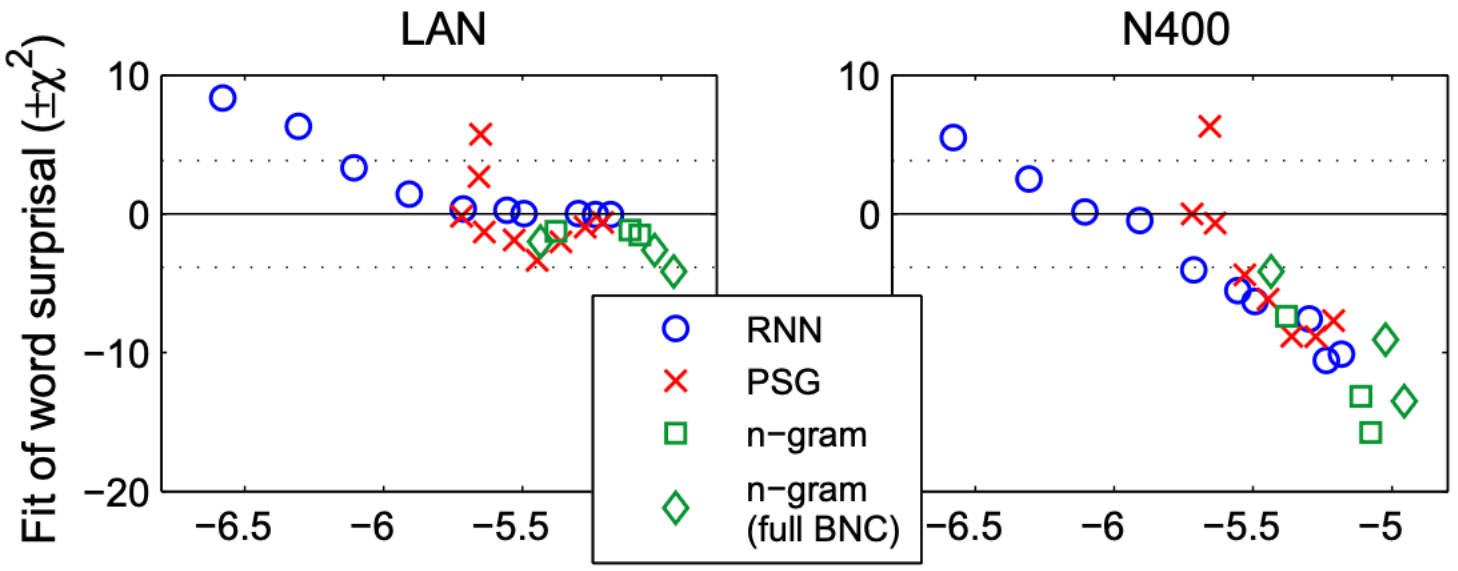
Gruteke Klein et al. (2024)



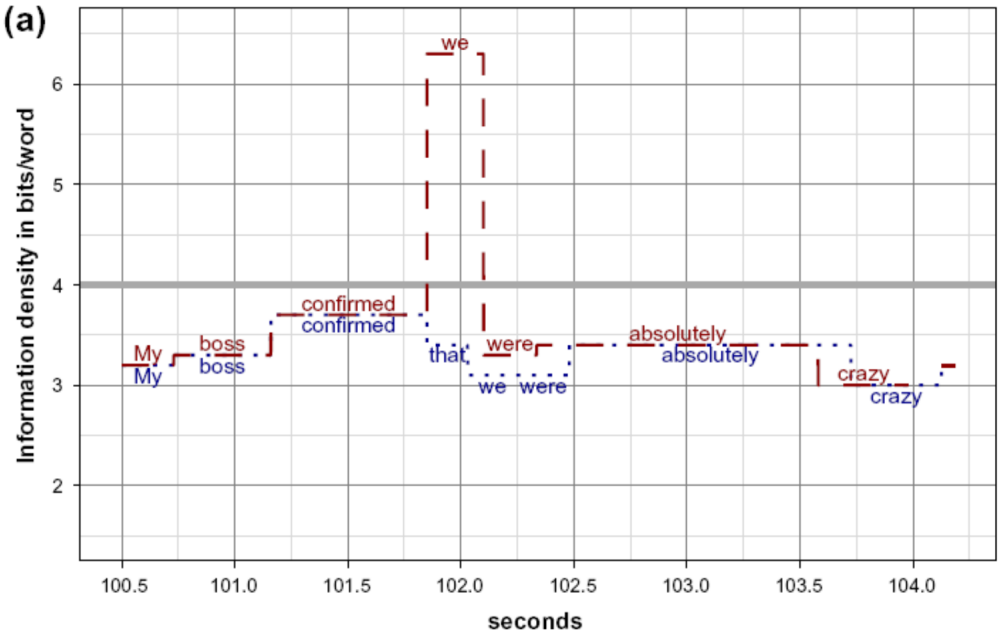
Source: Frank et al., 2015 http://dx.doi.org/10.1016/j.bandl.2014.10.006

EEG Evidence

- A comparison between the information measures and ERPs revealed a reliable correlation between N400 amplitude and word surprisal.



Evidence of UID at Corpus Level?



- Human speakers indeed tend to avoid the sudden changes in information in these very specific scenarios.
- How about a more generic trend at corpus level?

Which term is uniformly distributed?

$$H(X|Context) = H(X|C,L) = H(X|L) + I(X;C|L)$$

Approximated by surprisal

- Does $H(X|Context)$ keep relatively constant?
- UID => Entropy rate constancy (ERC) hypothesis.

Approximation due to the limited context

$$H(X|Context) = \underbrace{H(X|C,L)}_{\text{No}} = \underbrace{H(X|L)}_{\text{Yes}} - \underbrace{I(X;C|L)}_{\text{No}}$$

A sentence of *N* words, $w_1, w_2, \cdots, w_N$

$$H(X|L) \approx -\frac{1}{N}\sum_{k=1}^{N} \log p(w_k|w_{<k})$$ Average surprisal of all words in sentence



Estimated using LMs

Assuming UID/ERC is true

Constant ← $H(w|Context) = H(w|\boldsymbol{G},\boldsymbol{L}) = H(w|\boldsymbol{L}) - I(w;\boldsymbol{G}|\boldsymbol{L})$ Constant

Expected observation:

- $I(w_i; G\ |L)$ increases with i
- $I(w_i; G\ |L)$ decrease when topic shifts

Surprisal measures the
difficulty of understanding word w
Good speakers put hard words later (audience design)

Sentence 1: $\{w_1, w_2, \dots, w_6\}$

Sentence 2: $\{w_1, w_2, \dots, w_8\}$

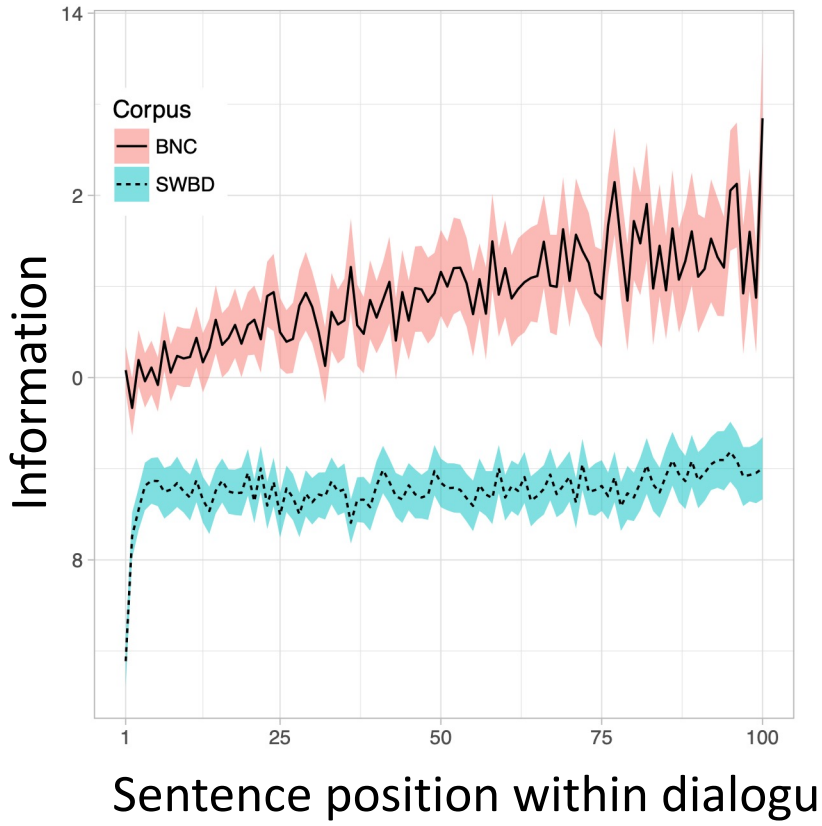
Sentence 2: $\{w_1, w_2, \dots, w_{10}\}$

…

Sentence *N*: $\{w_1, w_2, \dots w_{i-1}, w_i, \dots, \}$

Expected observation 2 confirmed (in dialogue)

- $H(X_i|L_i)$, surprisal, increases with i



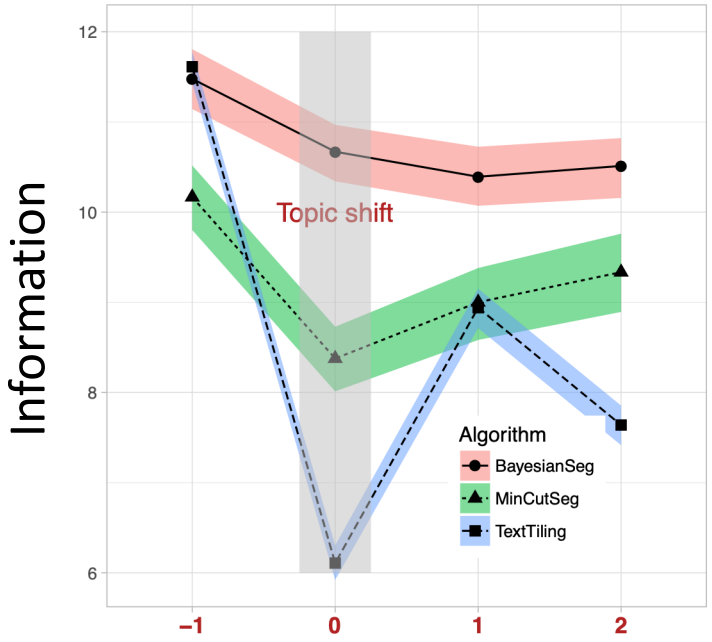
$$H(w|Context) = H(w|\boldsymbol{G}, \boldsymbol{L}) = H(w|\boldsymbol{L}) - I(w; \boldsymbol{G}|\boldsymbol{L})$$

Figure from Xu et al., 2016

Expected observation 2 confirmed (in dialogue)

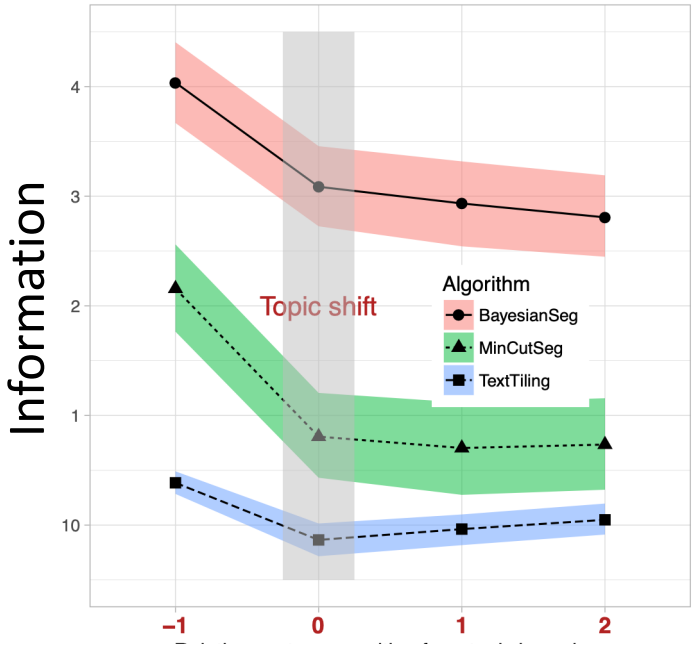
- Information suddenly **decrease** at topic boundaries.

Switchboard



Relative distance from boundary

BNC



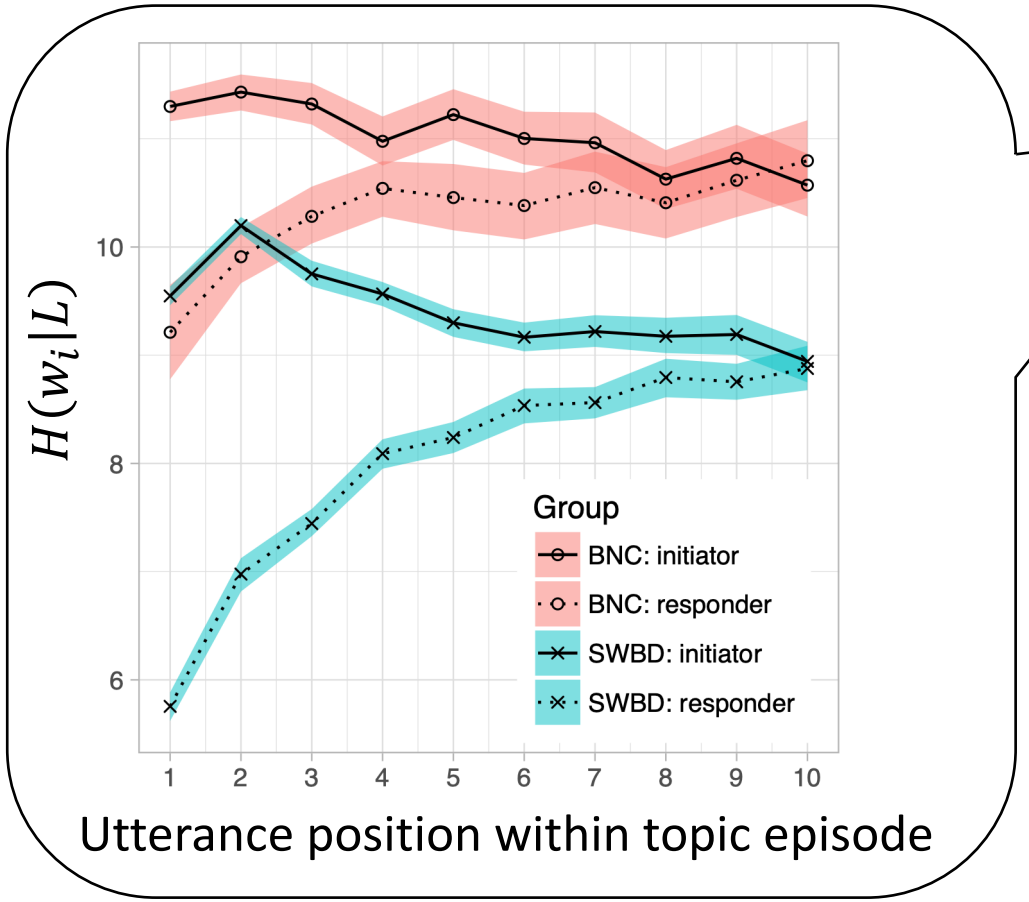
Relative distance from boundary

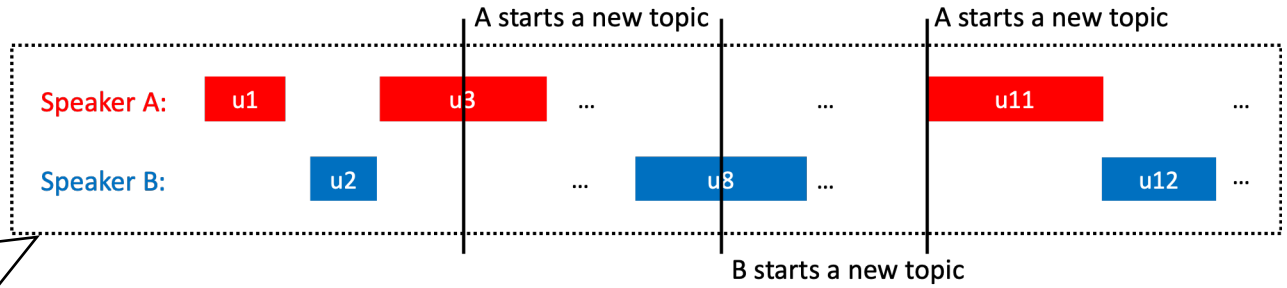
$$H(w|Context) = H(w|\boldsymbol{G}, \boldsymbol{L}) = H(w|\boldsymbol{L}) - I(w; \boldsymbol{G}|\boldsymbol{L})$$

Overview

- Language Model
- Cognitive Effort in Processing Language
- **Dialogue**

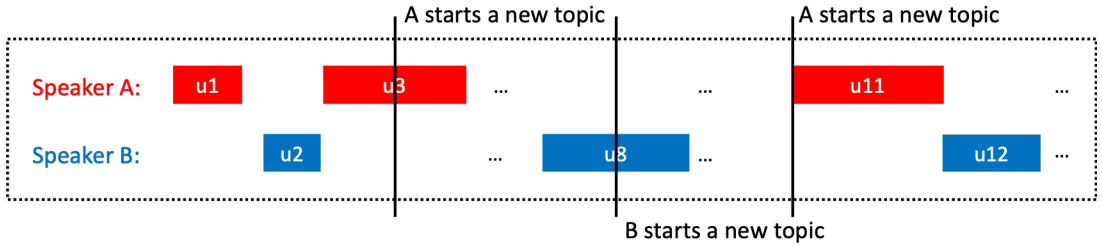
Surprisal $H(w|L)$ converges



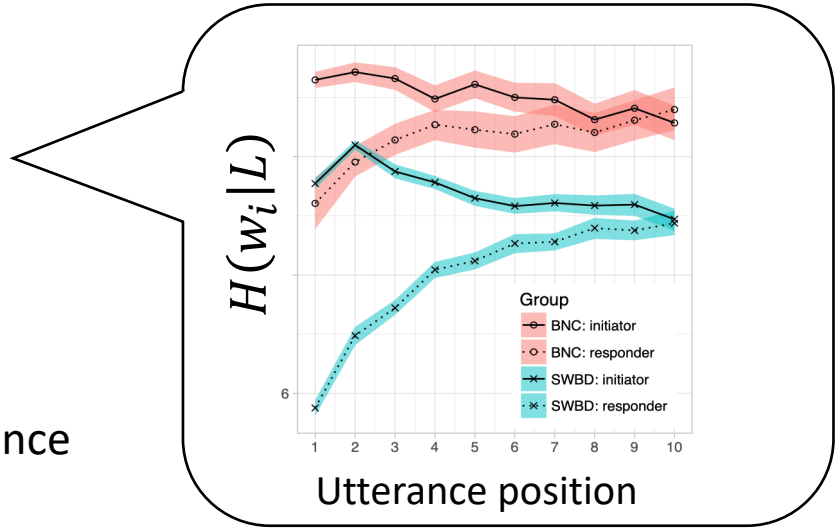


The local entropy (surprisal) of topic initiators and responders increases form a *convergence* pattern within topic episode

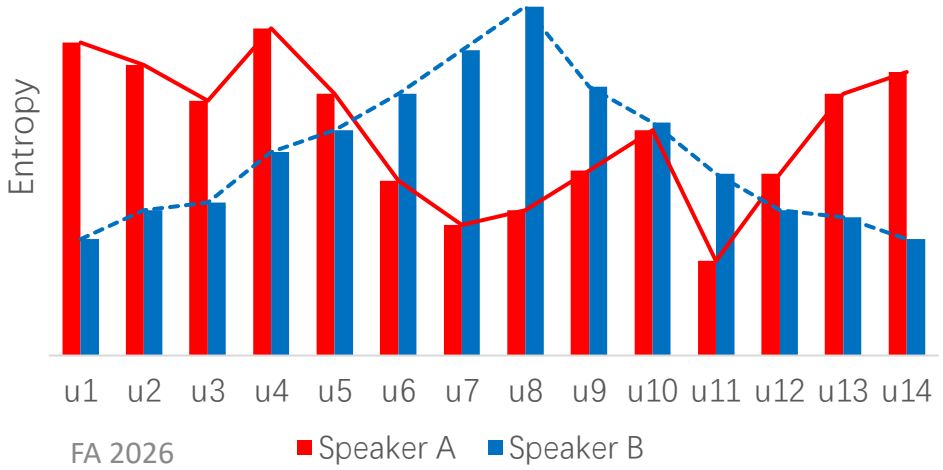
Periodicity of information in dialogue



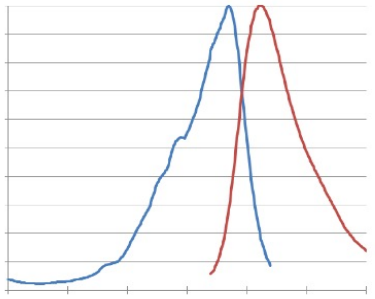
Topic shift occurs ***periodically*** → Aggregate → Convergence



Time domain



Frequency domain

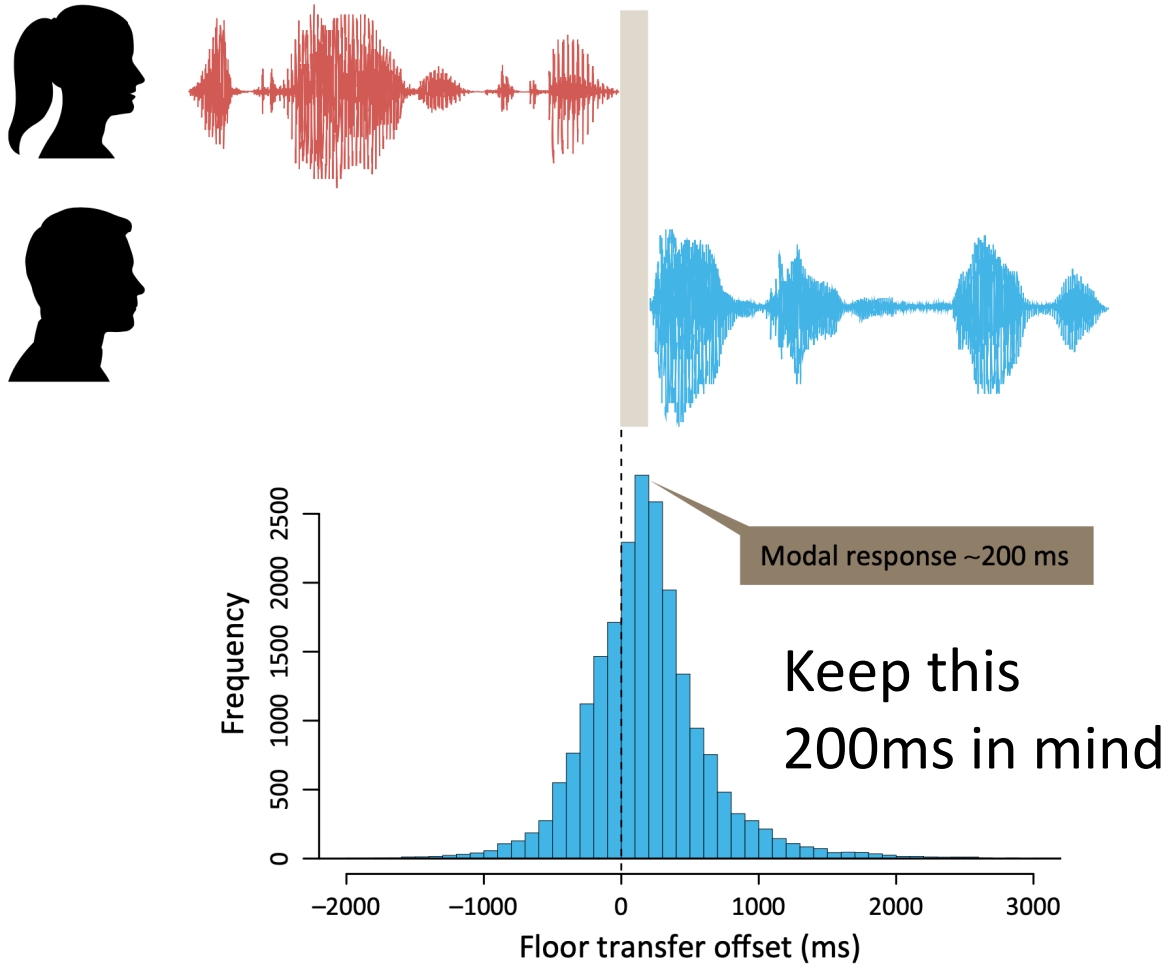


Features from frequency domain may reflect high-level dialogue properties

Turn-Taking 话轮转换

- Language is interactive, involving rapid turn-taking
- Turns are short and responses are remarkably rapid
- Part of a **universal infrastructure** for language

Responses in conversation are fast



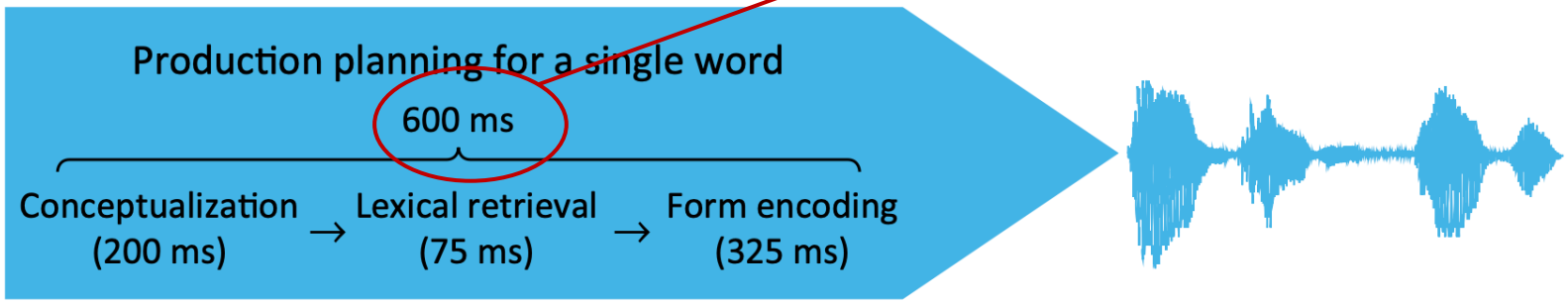
Keep this 200ms in mind

Source: Levinson, 2016

Turn-Taking

Latencies in production are **three fold**:

Longer than the **200ms** modal gap!
How do we explain this?



From **dense vectors** to one-hot

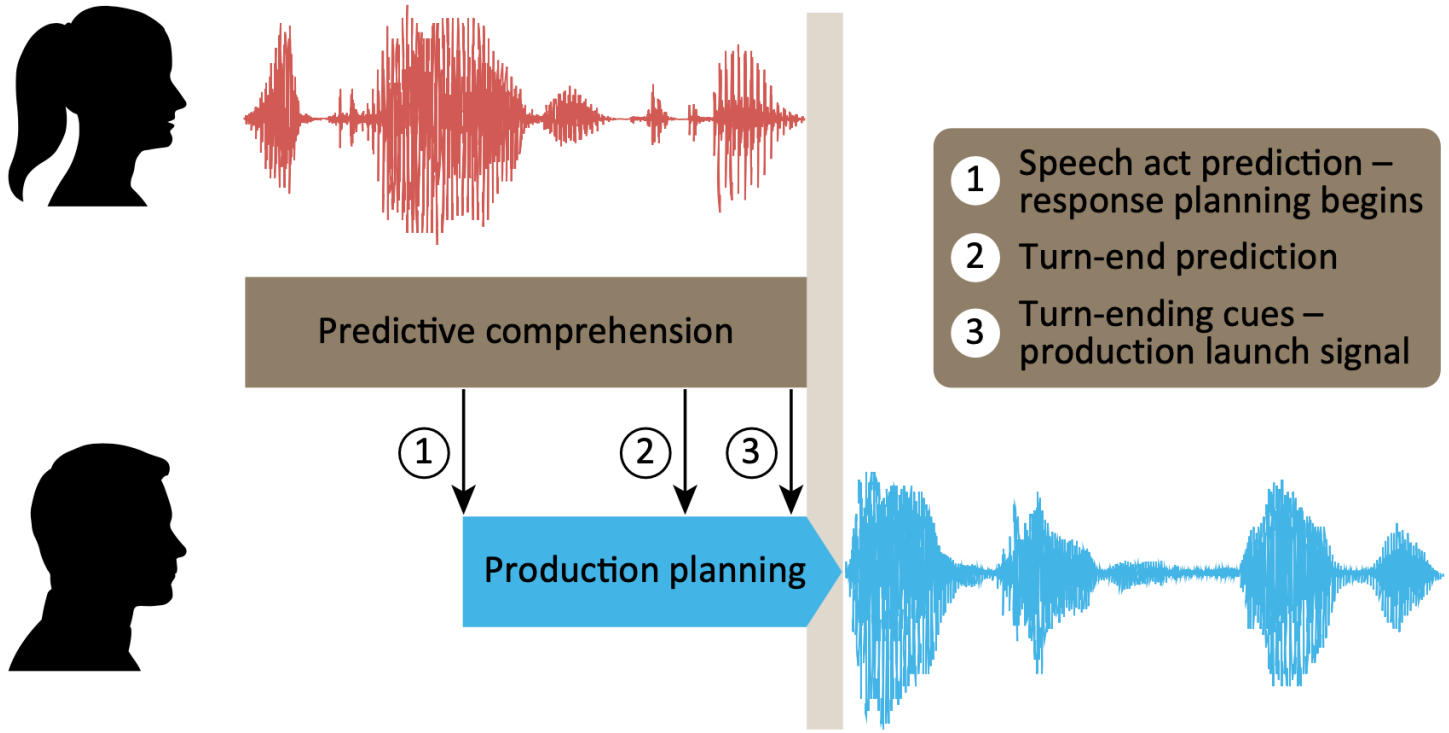
Lookup the vocabulary table

text-to-audio

Source: Levinson, 2016

Turn-Taking

Production of response must overlap with comprehension of the incoming turn.



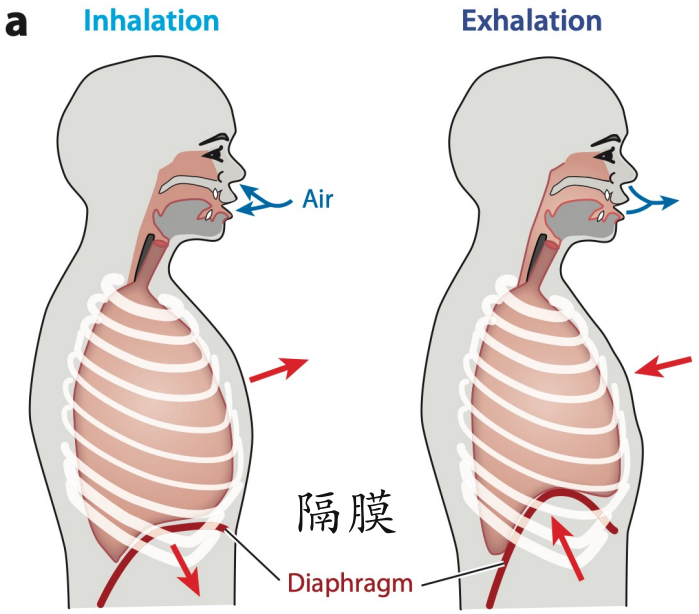
The slow production may be compensated for by predicting the **continuation** or **termination** of the incoming turn, and launching production early

Breathing is related to spoken language

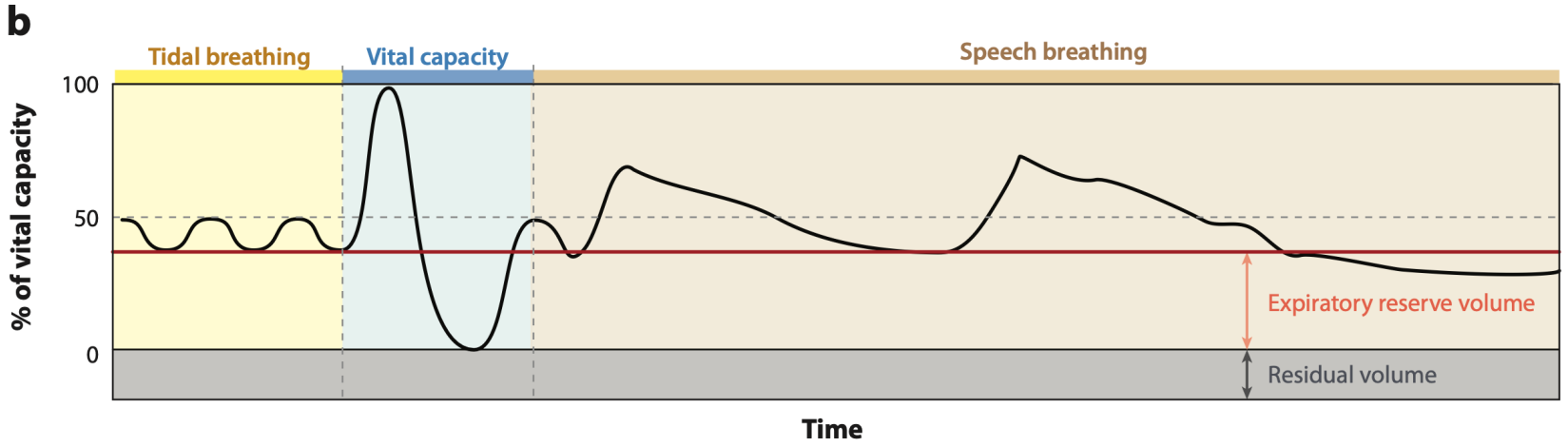
肺活量

Vital capacity: the maximum amount of air that can be breathed out after breathing in as much air as possible.

Speech breathing: involves more variable and deeper inhalations (更深的吸气) than tidal breathing



/ ˈdaɪəfræm /



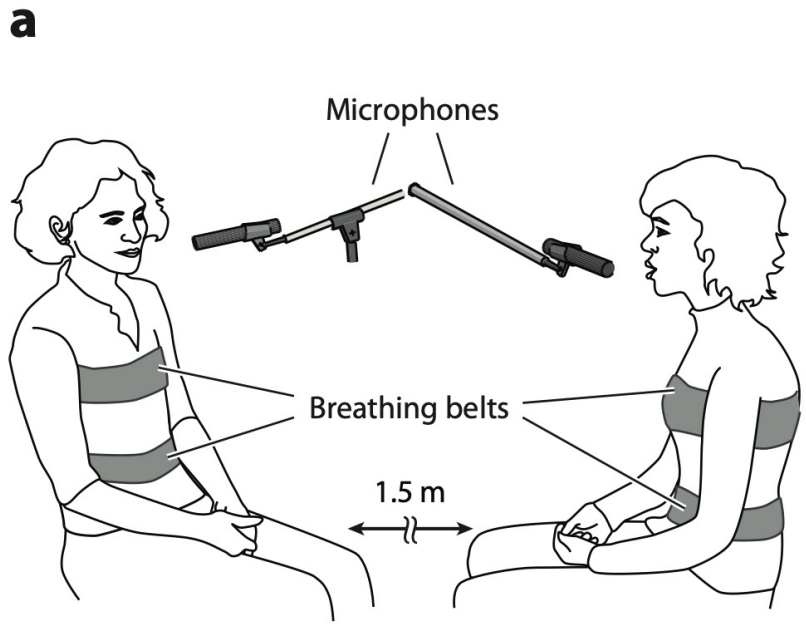
Tidal breathing: the term for inhaling and exhaling while you are relaxing

Speech breathing: short inhalation and a long exhalation

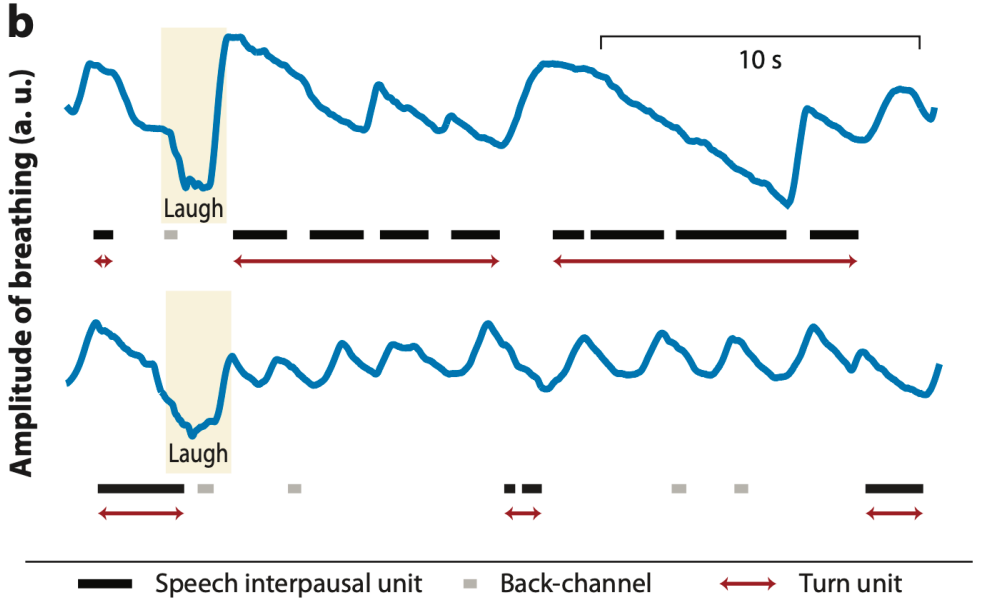
Source: Fuchs and Rochet-Capellan, 2020

Breathing during conversation/dialogue

- Techinique: Inductance plethysmography (电感体积描记法)



Breathing signals are recorded using breathing belts



Black bars indicate stretches of speech; red arrow indicates a speaker's complete turn

Breathing adapts to speech

- Inhalation (吸气) at **syntactic boundaries**
  - Rather than occurring randomly in the speech stream, inhalation very often coincides with constituent boundaries
- Anticipating the length of the upcoming sentence
  - Deeper inhalation for longer sentences than for shorter ones

